

Automating Code Generation from User Interface Prototypes

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Abstract—User-centred design is an iterative process that involves ideation, testing, and refinement to enhance usability, functionality, and overall user experience. However, there is still a need to code the final design. This transition from design to implementation is a time-consuming task. In this paper, we propose an approach and its supporting tool to generate a web application’s front-end code from a prototype of its user interface. Our goal is to generate code that can be easily integrated into the overall application development. The results demonstrate that the proposed tool can achieve visual fidelity comparable to that of existing commercial plugins, while producing higher-quality code that better reflects the behaviour specified in the prototype.

Index Terms—Web Development Automation, front-end frameworks, Atomic Design, Component-driven prototyping

I. INTRODUCTION

Web application development typically progresses through a number of phases, including analysis, prototyping, implementation, testing, and maintenance. Within this process, User Interface (UI) development represents a significant portion of the workload, be it during the initial creation of the product or subsequent improvements [1]–[3]. Frontend development can be particularly time-consuming due to the complexity of tasks such as handling user interactions, managing application state, and integrating with backend services.

Despite the availability of numerous frameworks, component libraries, and development tools designed to assist developers, UI development continues to constitute a substantial and essential part of the development effort. A core challenge is the dual perspective involved in building user interfaces: on one hand, designers approach the interface from a User-Centred Design (UCD) perspective, focusing on usability, design ideation, and validation with end users; on the other hand, developers focus on implementing the interface in a reliable and maintainable form. These two perspectives, although complementary in goal, operate with different assumptions and artefacts. UCD emphasises early-stage exploration, informal prototypes, and iterative feedback loops to ensure that the interface meets user needs. Developers focus on translating design into code, which can be time-consuming and error-prone, especially when requirements are constantly evolving.

One approach that has been proposed to help bridge the gap between design and implementation is Model-Based User

Interface Development (MBUID) [4]. The goal is to reduce the manual work involved in UI development and improve consistency by transforming high-level abstract models that describe the UI into implementation code. However, because it relies on formal models rather than informal sketches or prototypes, MBUID does not easily fit into the iterative, user-driven nature of UCD, which hinders its adoption.

Advances in prototyping tools have narrowed the gap between UCD and implementation. Tools like Figma or Sketch support structured, machine-readable exports that go beyond static mockups. This allows prototypes to serve not only as communication artefacts but also as potential inputs for automated development workflows. In this paper, we propose a tool (FIGMA2VUEJS) that follows an MBUID approach to reduce the disconnect between UCD and implementation. By viewing user interface prototypes as formal interface models, our tool supports a transformation pipeline that bridges informal design with structured development. This allows developers to benefit from automation while preserving the flexibility and expressiveness valued in user-centred design processes.

II. RELATED WORK

Tools have been developed to ease the design and prototyping of user interfaces. Examples include Figma, Sketch, UXPin, Moqups, and others [5]. Amongst these options, Figma has gained popularity due to its capabilities, such as real-time collaboration, cross-platform compatibility, and an extensive library of components [5]. Figma prototypes are organised into frames (also known as *artboards*). Design elements can be grouped into components, which may have variants to reflect different states or interactions. Transitions between frames can be defined to simulate user interactions. The tool supports plugins that enhance various aspects of the design workflow, such as layout, content, accessibility, and code generation, thereby reducing the gap between design and implementation.

One of the most popular code generation plugins is the Anima plugin [6], which uses machine learning to map design components to their equivalent code. Anima can generate projects for various frontend frameworks such as Vue.js and React, making it a useful starting point for representing the visual structure of the UI. However, the generated code primarily focuses on visual presentation and lacks functional implementation (i.e., behaviour logic). For instance, it lacks

José C. Campos was supported by the *Fundação para a Ciência e a Tecnologia* (FCT), Portugal, under Contract FCT/Mobility/134021011/2024-25.

code for opening and closing overlay elements or managing the state of component variants. Furthermore, the generated UI is not fully responsive, and the code presents poor practices such as absolute positioning. Other drawbacks include issues related to interpreting the referential coordinates from Figma *artboards*. In addition, the effectiveness of these plugins is influenced by the organisation of the source code generated, the selected builder, and the version of the framework used in the project. The plugin integrates Generative AI features that enable developers to adjust the code in relation to the design, UI logic, or additional functionalities using natural language prompts [7]. However, these techniques may generate code that does not align with the designer's intended behaviour, potentially delaying the development process.

Besides Anima, many other LLM-based solutions have been proposed with the aim of converting prototypes to web apps. Two examples are Figma Make [8] and Vercel's V0 [9]. In these approaches, it is important to provide well-crafted and descriptive prompts in order to give context to the LLM. This helps the LLM correctly interpret which prototype elements correspond to pages in the web application or what behaviour is expected for navigation actions. This aspect represents a disadvantage since it implies replicating, in the textual prompts, information that is already in the prototypes. In cases where the programmer wants to adjust the generated code, obtain code related to page routing, or integrate UI libraries, it will be necessary to provide that extra information. Regarding UI layout, this method is not entirely accurate, especially in the initial iterations, necessitating multiple rounds of refinement and increasingly detailed prompts to achieve acceptable results. Compared to the first approach of code generation through plugins, Vercel's V0 overall shows lower code quality in terms of interaction logic and UI layout.

In recent years, low-code and no-code development platforms have gained significant traction in the software industry [10]. These platforms aim to reduce the amount of hand-written code and abstract away low-level concerns such as database configuration, ORM mapping, and integration with external services. Typically, they provide visual development environments equipped with pre-built user interface components, allowing developers and designers to assemble applications by reusing these standardised elements across projects. This approach builds on component-oriented development, which offers several advantages. By relying on well-defined, reusable components, these platforms can automate aspects of UI design and source code generation while improving maintainability. Since component structures are often immutable and consistent, updates and modifications become easier to manage. One limitation, however, is the loss of design flexibility. Users are constrained to the components and layouts offered by the platform. This can become a barrier in projects that require a high degree of customisation or originality in the interface design. As a result, while low-code solutions are well-suited for the rapid development of standardised applications, they may not be ideal for projects with unique or complex UI requirements.

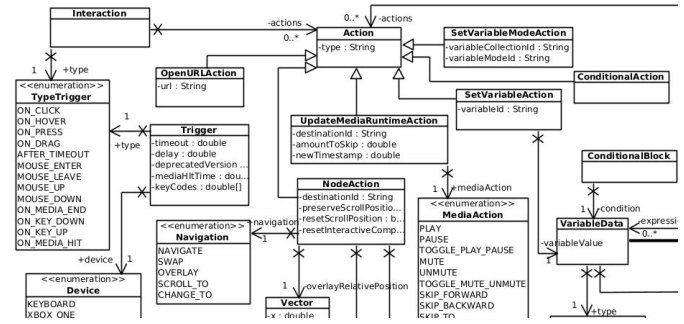


Fig. 1. Metamodel of Figma prototypes (Action & Interaction excerpt)

III. PROPOSED APPROACH

Our approach aims to combine the flexibility of prototyping with the automated code generation of MBUID. The strategy to achieve this consists of viewing the prototype as the user interface model from which the code can be derived. This prototype-derived model is then progressively refined into a model of the implementation that supports code generation for a target UI framework.

Metamodels for the prototypes and the target UI framework are needed to support the refinement process. We will focus on Figma prototypes and the Vue.js framework (version 3). Obtaining information about the prototype is feasible through the Figma API, which makes it possible to extract information about the prototype and its nodes. A Figma prototype is characterised by different types of nodes (Frames, Sections, Instances, Components, Rectangles, etc.) with their associated styling (sizes, colours, strokes, effects, etc.), organised in a tree. The prototype also has information related to behaviour, such as user interactions and actions. Figure 1 presents an excerpt of the metamodel, focusing on interactions and actions.

A Vue.js project's component hierarchy is again organised in a tree, in which the App is the root and the components are the leaves. Each component will have a template section with a set of Vue directives, a script section, and a style section. The project will also have a router that handles navigation between pages, a set of plugins, and a store to manage the application's state. Figure 2 presents the metamodel.

The approach leverages component-oriented prototyping based on the principles of Atomic Design [11]. Atomic Design makes an analogy between HTML elements and chemical elements. It allows us to have a design system with elements that interact with each other to form components with well-defined structure, styling, behaviour, and logic. This will later facilitate the process of generating the corresponding code since elements in the design system are known in advance. The component-based approach seeks not only to simplify the design process but also the implementation phase. Generating the code of these components will be done using UI libraries such as Vuetify and PrimeVue, resulting in code with greater modularity and adaptability for future adjustments by developers. When elements are composed directly in the prototype to create new components, the analysis will make use of

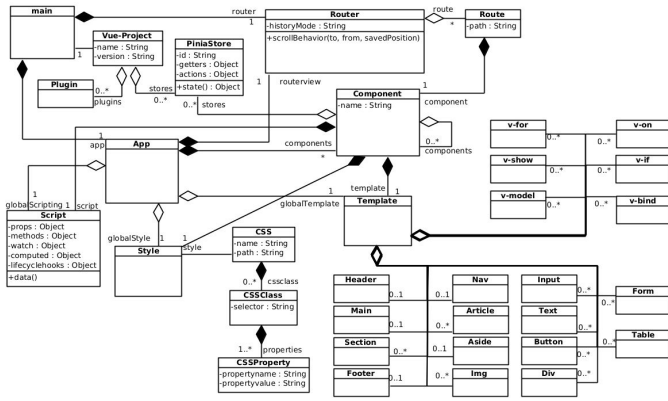


Fig. 2. Metamodel of a Vue.js project

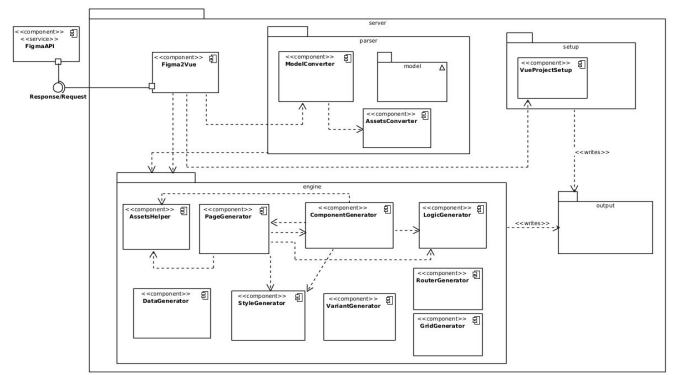


Fig. 3. Component Diagram (server-side)

containment information relations (both structural, through the elements' containment hierarchy, and graphical) to identify and process them. Regarding layout, the CSS grid system will be used. Through analysis of the position of the elements in the prototype, the best positioning of the elements in the grid will be determined.

IV. IMPLEMENTATION

In this section, we discuss the implementation of the proposed tool (FIGMA2VUEJS).

A. Architecture

As explained above, a model derived from a Figma prototype will be transformed into a Vue project model. In this process, an intermediate model will contain the information extracted from the prototype and will be structurally organised to link the two models.

Relevant entities in this intermediate model include: *Mpage*, *PageItem*, *Melement*, *Mcomponent*, *InteractionElement* and *ElementAction*. All top-level frames in the prototype with “#Page” annotation in their name will correspond to an *Mpage* entity. This entity will originate a Vue page with the relevant contents. The entities *ContainerElement*, *ShapeElement*, *TextElement*, and *ImageElement* will be subentities of *Melement*. The *PageItem* holds a self-referencing list, creating a tree structure, in the same way as elements can be nested in a prototype. This entity will be a generalization of *Melement* and *Mcomponent*. This supports the representation of components defined directly in the prototype. If a design system is used, its components (which are predefined) are represented by subclasses of the *Mcomponent* entity. This organisation provides flexibility as both prototype-defined components and design system components can be represented. The latter can then be mapped to implementation components from Vue UI libraries (e.g., from Vuetify). The former will have to be analysed, and code will be generated to represent them.

The *InteractionElement* entity will correspond to the interactions that users can have with a given element of the prototype. Among others, this entity is characterised by a trigger action (e.g., *onclick*, *onhover*, *ondrag*, etc.) and a response,

such as navigation, opening overlay elements, changing variant states, and scrolling events, among others. The *ElementAction* entity will be the superclass of trigger response actions.

Regarding the general architecture adopted for FIGMA2VUEJS, a division by architectural components was followed (Figure 3), with a view of separating blocks by their responsibilities. The Figma2Vuejs component is the tool's entry point. This is where the data will be fetched from the selected prototype through the Figma API. At startup, it can receive information regarding the grid system to apply to the Vue pages. The ModelConverter component is where the data obtained from the Figma API will be parsed, and where the intermediate model described above will be built and stored. The AssetsConverter is where sizes, colours, texts, images, and other properties will be extracted to build the components that will later be converted into Vue UI library components. The setup package will contain a VueProjectSetup component that will create the Vue project with the name associated with the Figma prototype, and inject all the necessary dependencies and components from the design system present in the prototype. The engine package is where most of the generation and rule transformations will be done. The PageGenerator and ComponentGenerator components process and create the Vue pages and UI components, respectively. The StyleGenerator maps the elements' properties to CSS rules. The LogicGenerator is responsible for creating functions and JavaScript code associated with different Vue hooks, depending on the element's behaviour. The RouterGenerator generates code that is responsible for forwarding each route to a Vue page. The VariantGenerator generates dynamic components, which change their state whenever a user-triggered action is detected. Lastly, the DataGenerator is responsible for creating data objects to be displayed on the presentation layer.

B. Major Challenges

The generation of a Vue project from an intermediate model raises challenges that are discussed in this section.

1) *Challenge 1 – Automating the setup of a Vue project:* Setting up a Vue project involves multiple manual steps:

initialising the project, configuring routing and state management, installing dependencies, and removing boilerplate, which can be time-consuming and error-prone when done repeatedly or at scale. The strategy followed consists of using Python's "*subprocess*" module, which allows us to run bash or shell commands in the Python code itself, which are then executed by newly created processes. Functions were also created to manage and inject the UI library components in *main.js*, and to set a toast service store to explore the reactivity of elements such as Forms and Sliders when data is submitted. After obtaining the Vue project, additional processing is done to add global styling and to remove files with boilerplate code that isn't necessary. In this block, the installation of project dependencies is also automated by creating a shell script file with the *npm install* and *run* commands to execute the project.

2) Challenge 2 – Element positioning and grid system:

Translating absolute positioning from Figma prototypes into a responsive and maintainable CSS grid layout poses a significant challenge due to differences in coordinate systems, nesting behaviour, and the need for layout normalisation. The strategy followed aims to provide the user with the option of choosing the desired division for the web pages. By default, each page is divided into a grid with 64 columns and 128 lines, where row height is dynamically calculated based on the page's height in the prototype, to allow for vertical growth. Figma's API coordinate system uses the upper-left corner of the *artboard* as its reference point. Therefore, the coordinates of each element inside a prototype's page need to be normalised to be relative to the page's origin. Firstly, we adjust every node to consider the coordinates of its parent node in the hierarchy as its reference point. Then, a translation of each page to the point (0,0) of the *artboard* is performed. Thus, based on the length and width of a node and its parent, it's possible to calculate the *grid-column-start*, *grid-column-end*, *grid-row-start*, and *grid-row-end* properties for each element. The algorithm will be applied recursively so that each parent node has an associated subgrid, ensuring consistent layout behaviour across all hierarchy levels.

3) Challenge 3 – Transformation rules: Defining transformation rules that convert abstract design nodes into meaningful, semantically appropriate HTML elements is a complex task, particularly when accounting for variations in node types, user intent, and interactivity. To manage this, the tree structure that represents each prototype's page is iterated over to apply the appropriate transformations for each node type. Each class representing a prototype node has an associated object corresponding to the node's style. One of the attributes of these classes is the 'tag'. If the designer appends the *#tag_name* annotation to a node's name, its value is used to guide the transformation and defines the HTML tag to be generated. Otherwise, default mappings are applied: ImageElement objects will have the 'img' tag, TextElements will have the 'p' or 'a' tags (depending on whether they have a navigation action), ContainerElements and ShapeElements will have the 'div' tag. Additionally, each Figma node has a unique ID, which makes it easy to manage CSS selectors and assign properties like font

sizes, gradients, colors, etc., to their respective values.

4) Challenge 4 – Design system & Vue plugins: Integrating a custom design system with existing Vue UI libraries presents a challenge due to the need to accurately map abstract design components to concrete implementations while preserving intended behaviour and appearance. The strategy for converting components from the design system into functional components of PrimeVue and Vuetify begins by reserving the following names: *InputSearch*, *DatePicker*, *Dropdown*, *Read-OnlyRating*, *InteractiveRating*, *Paginator*, *Form*, *Checkbox*, *Menu*, *Slider*, and *Table*. When these elements are identified in the prototype, a parsing step is performed over them, and relevant properties of each type of component are saved (e.g., list of items, colours, input text, etc.). Depending on the components detected, the algorithm maintains a dictionary of dependencies to update the content of the plugin files dynamically. The components from our design system are mapped to components from either PrimeVue or Vuetify, ensuring seamless integration between the custom design and the selected Vue UI libraries.

5) Challenge 5 – Extracting images and icons: Extracting image and icon assets from Figma prototypes introduces challenges related to network dependency, format consistency, and efficient batching of API requests. Since Figma files are stored on Amazon AWS, it is possible to select specific nodes from the prototype file and download them in a given format, such as .svg or .png. To manage this process, two global lists are initialised to store the IDs from nodes of vector type and of rectangle type, respectively. By storing these resources locally, generated Vue code becomes independent of external services, unlike the code obtained from the Anima plugin. Furthermore, the algorithm was adapted to make API calls in batches of size 50, in order to avoid large request payloads.

6) Challenge 6. – Overlay elements: Managing overlay elements in a prototype is challenging due to the need to accurately position overlays relative to trigger elements and to maintain correct hierarchical relationships within the page's DOM structure. From the element that triggers the action, it is possible to extract the offset coordinates (rx, ry) where the overlay should appear. For elements such as Frames or other first-level artboard nodes involved in hover interactions, it is necessary to check the location of the trigger element within the page's DOM hierarchy. Suppose the trigger element is contained within a node belonging to one of the pages. In that case, the algorithm dynamically inserts the overlay element into the trigger element's parent (the immediate hierarchical ancestor). In cases where the overlay overlaps the trigger element, the involved elements are assigned a hover property: the overlay starts with an opacity of 0, and transitions to an opacity of 1 on hover. At the same time, the trigger element transitions from opacity 1 to 0. If the overlay overlaps any component, the component's structure is recursively modified to include the overlay as a child of the relevant parent node. Furthermore, the algorithm builds the path from the trigger element to the overlay element, which is necessary for managing signal emissions in opening and closing actions.

7) *Challenge 7 – Overlapping on shape elements:* Shape nodes (geometric objects in Figma prototypes, such as *rectangles*, *ellipses*, *stars*, etc.) do not support nesting of child nodes, even though we can overlap elements on top of them visually (depending on the order of layers in the prototype). If a node is within the spatial boundaries of a shape and appears after it in the layer list, the shape will visually overlap that node; if it appears before, the opposite occurs. The algorithm traverses the DOM structure of each page, and, for each pair of elements, if one is a shape node with a lower index in the elements list, and the grid boundaries of the other element are fully contained within it, the second element is added as a child of the *shape*, which is configured to have a grid display. This strategy is applied recursively to all child nodes, across every page of the prototype.

8) *Challenge 8 – Variants conversion:* Variants provide a way to define multiple configurations or states for a single component. The modelling strategy begins with a top-level dynamic component, which serves as a container for all variants. Each variant is converted into its own state component, representing a specific configuration. These state components are managed by the dynamic container, which uses a computed property to determine the current state based on user interaction. The logic for handling state changes involves passing the selected variant and the relevant selectors (keys or properties associated with each state component) as props. This enables the dynamic component to apply the appropriate properties and render the correct state. For each dynamic component, its state components and their corresponding selectors are stored.

9) *Challenge 9 – Data generation:* Extracting structured data from design elements is challenging due to the need to infer semantic meaning and group related fields without explicit data models. In order to represent a data entity, a “#data” tag must be appended to the names of the relevant nodes. When the algorithm finds a “#data” node, it performs a recursive parsing of the node to extract attribute–value pairs. Nodes in the same container will be part of the same data list. If a data entity is not enclosed in any container, the entity is added as an individual object in the page’s data property. When generating a page or component, the ID of the corresponding node is needed to perform data interpolation. To achieve this, a dictionary is passed through props, allowing the entity’s values to be dynamically interpolated into the component template.

10) *Challenge 10 – Behavior’s logic code generation:* Mapping user interactions from Figma prototypes to appropriate Vue event handlers requires supporting a variety of interactions and automatically generating meaningful and maintainable logic. Figma interactions may map directly to some Vue event-handlers (e.g. *on_click* to *@click*), but some require custom logic (e.g. event listeners are used for *on_key_down*; *setTimeout* function is used for *after_timeout* on *created* hook). A dictionary data structure was created, where each key represents a Vue hook and maps to a list of method names and lines of JavaScript code that should be executed when the event is triggered. Different methods were created depending on the type of action. For example, in Navigation, the *\$router*

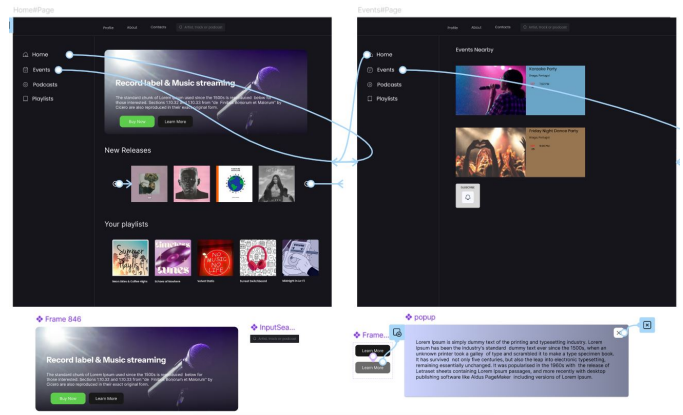


Fig. 4. Use case prototype made in Figma

property is used, in Scroll, the *scrollIntoView* method is used, in Overlay, the path of the signal emissions is tracked, and the visibility of the elements involved is changed depending on the event, etc. Additionally, well-defined design system components allow for automated behaviour and logic. For instance, when the *InteractiveRating* component is detected we import the *VRating* from *Vuetify*, when a *Form* component is detected, validation methods are created, the *Slider* component is configured to display a toast message showing the current value whenever it changes, for the *Dropdown* and *Checkbox* components, data is pre-populated within the mounted lifecycle hook, simplifying future integration with APIs, etc.

V. RESULTS

To validate the tool, a set of 36 prototypes (some purpose-built, others imported from open-source projects) was converted to code with both our tool and the *Anima* plugin. In the imported prototypes, modifications were made in the nodes’ names, and components from our design system were inserted. The prototypes were divided into five categories of incremental complexity, ranging from basic nodes to advanced elements such as components, design systems, variants, and the behaviors associated with them.

Figure 4 illustrates one of the prototypes used. This prototype, a generic music streaming platform, has two pages, one variant component, one generic component that contains the variant, one component from our design system (*InputSearch*), and one overlay component. The prototype features: page navigation; scroll and draggable elements; interaction with modal, input, and button components; style representation, such as colour gradients, text fonts, sizes, shapes, images, etc.; reusable components; display of data; and state changes through variants.

A. Visual Fidelity

To analyse the outcomes (cf. Figures 5 and 6), metrics were calculated to determine any visual differences between the generated web UI’s and the Figma prototypes. First, a pre-trained Convolutional Neural Network (CNN) based on the *ResNet50* model was used to evaluate the cosine similarity

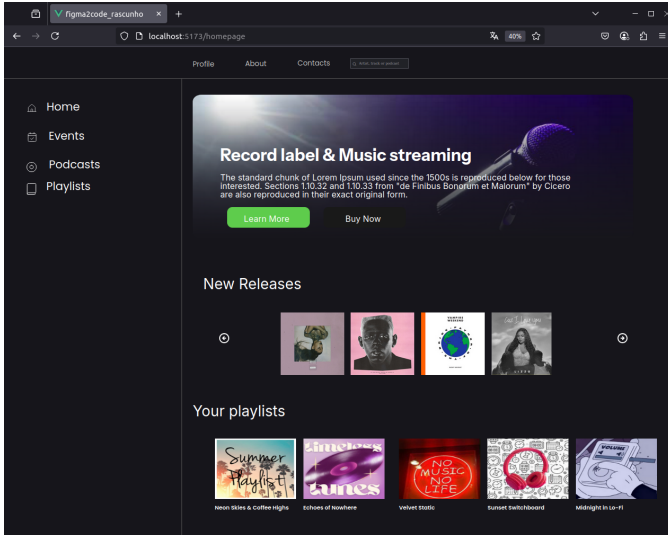


Fig. 5. Conversion of the use case of Figure 4

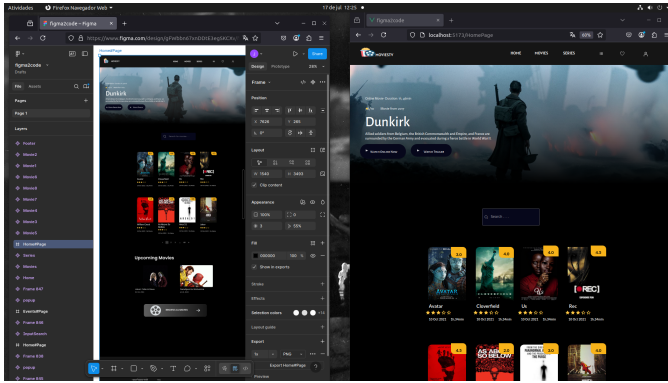


Fig. 6. A complex prototype (left) and its FIGMA2VUEJS web output (right)

and the Euclidean distance between the set of generated web UI's and the set of prototypes. When an image is passed through the CNN model layers, filters (matrices that were learnt from training) are applied to the image, which generates other images with different features detected from the original image. As each filter identifies and highlights a specific pattern from the image, it produces a feature map, in which the tensor values vary depending on whether a feature was detected more in a region or not. We extracted these feature maps from the intermediate layers of the model to calculate the activation differences between the two inputs (images generated with our tool and images from the prototype) and visualise the low and high-level differences through heatmaps. This allowed us to identify which types of elements differ the most and which are identical or nearly identical between the two sets.

The other metric considered is the Structural Similarity Index Measure (SSIM), which provides a perceptual comparison based on structural interdependencies between pixels [12]. We measured the SSIM between the web UI's and the prototypes using Python's module `structural_similarity` from `skimage.metrics`, and analysed how accurate the images were

and where they diverged using the OpenCV library.

For this purpose, 34 images of web pages generated with our tool and their respective prototypes on Figma were exported to .png and .jpg formats. The ResNet50 model was loaded with the pre-trained weights of the ImageNet dataset. The top layer, which outputs the final classification, was removed, and the average layer pooling was selected to compute the average of the feature maps. The two sets were preprocessed in batches and passed to the model. The resulting embeddings were extracted, and the cosine similarity was calculated through the dot product of each row (which represents one of the images) of the normalised embeddings. In order to calculate the Euclidean distance, we used Python's `numpy.linalg.norm` module with `axis=1` to get a list of distances for each pair of images. For L2-normalised embeddings, the cosine similarity ranges from -1 (opposite direction) to 1 (same direction), and the Euclidean distance ranges from 2 (opposite) to 0 (identical).

TABLE I
COSINE SIMILARITY AND EUCLIDEAN DISTANCE METRICS

N=17 pairs	cosine similarity	euclidean distance
average	0.96	0.27
least similar	0.88 (pair 1)	0.49 (pair 1)
most similar	0.99 (pair 15)	0.17 (pair 15)

Overall, the generated UI and the respective visual Figma prototypes are very similar, resulting in an average cosine similarity of 0.96 and an average Euclidean distance of 0.27. Even though the most deviated pair scored 0.88 for cosine similarity and 0.49 for Euclidean distance, the differences are not too significant. In order to understand where they diverge, we built a secondary model that outputs some of the intermediate layers from the previous model. Three layers were selected: `conv1_relu`, `conv2_block3_out`, and `conv3_block4_out`, which capture low-level (eg, textures) to higher-level features (eg, structural layout). Next, the prototype embeddings and the generated UI embeddings were extracted from those selected intermediate layers, which resulted in 4D tensors of shape (batch size, height, width, channels). Finally, for each pair of corresponding feature maps, the absolute difference was computed, and the sum across all channels was computed at each spatial location. This produced a 2D heatmap in which each position indicates the magnitude of the feature difference between the pair of webUI and prototype.

The values for each spatial location on the images are represented with colours that vary in intensity, ranging from black, red, yellow, and white (black representing almost identical regions, and white representing regions with more discrepancies). As seen in Figure 7 (top), the lightest region coincides with a FORM component of our design system. This occurs since we are converting these types of components into blocks from which only key features are being extracted. In this specific case, we are extracting the number of inputs, placeholders, the form position, and colours from the input, text, and button. However, other features, such as spacing between inputs, the width and height of each input, text fonts,

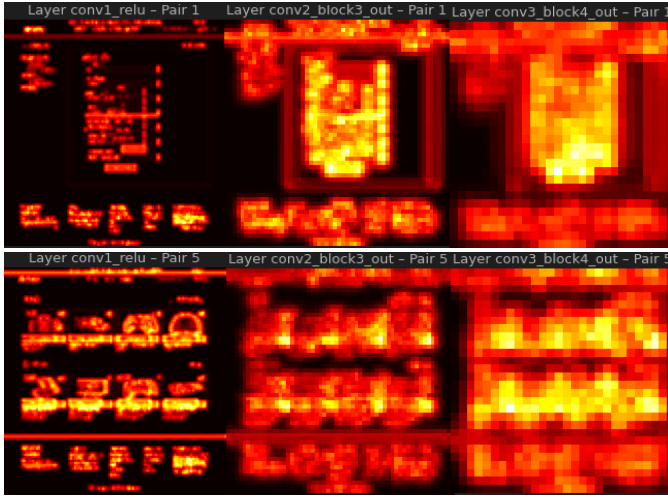


Fig. 7. Feature map differences in 3 different layers of the model: pair 1 (top), pair 5 (bottom)

etc., are aspects that need to be taken into consideration when overwriting the styling of these components from UI libraries. Meanwhile, in Figure 7 (bottom), the lighter regions correspond to shape elements such as rectangles, ellipses, etc., that are being implemented with the clip-path property, and sometimes the elements contained are being cropped and a bit unformatted due to the grid-system allocation.

TABLE II
COSINE SIMILARITY, EUCLIDEAN DISTANCE, AND SSIM

	Anima	FIGMA2VUEJS
cosine similarity	0.97	0.96
euclidean distance	0.23	0.27
ssim	0.79	0.81

On average, Anima performed slightly better at maintaining geometric fidelity in the conversion to Vue code. In some cases, layouts were significantly impacted by Anima's approach to implementing component states and overlay elements. In other cases, pages were not created and therefore did not affect the above metrics. Although the vector similarity is slightly lower, our tool has the advantage of representing the UI in a grid system, which is a good design practice for screen adaptability. In Anima, as that approach was not followed, we had to adjust the size of the generated page to the screen in order to screenshot the viewport and calculate the metrics; otherwise, that would have an impact on the metrics. However, our tool performs slightly better on the SSIM metric (0.81 vs. 0.79 for Anima), which ranges from -1 to 1 (1 indicating perfect similarity between two images and negative values indicating dissimilarity). SSIM indicates how similar the prototype image and the respective web UI image are from a human visual point of view (based on contrast, luminance, and structure). Although our tool achieves a slightly higher SSIM score, given the perceptual nature of SSIM and the small absolute difference, the two approaches can be considered approximately equivalent in terms of visual

fidelity from a human point of view. For replicability purposes, the evaluation code is available in a Jupyter Notebook at [figma2vuejs-evaluation](#).

B. Code Analysis

As a method of validation, the Vue code generated from our tool was evaluated using SonarQube. The evaluation of the code quality was based on the following metrics: security, reliability, maintainability, size & comments, and complexity.

A complete code review found no security vulnerabilities or potentially dangerous patterns (score A). Among the 36 projects, the code holds an A reliability rating, though 25 Medium-level issues remain open. These issues are related to the fact that SonarQube does not detect components imported from Vue UI libraries, contrary to native elements. Therefore, a lack of matching control over *input* elements was reported, even though an ID was associated with the component used to implement them (InputText from the PrimeVue library). In the maintainability metric, an A rating was achieved, with 72 code smells detected across the 36 Vue projects (average of 2 per project). The total technical debt is approximately 2 hours and 52 minutes, with all issues classified as low complexity. These issues are related to duplicated CSS selectors in variant components and duplicated CSS selectors from component instances.

Regarding the size & comments metric, the 36 projects contained 178,572 lines of code (LOC), 1255 files, 1423 functions, and 171 comment lines. As the current amount of comment lines represents only 0.1% of the overall code, further effort is needed to generate docstrings for more complex functions, provide additional comments indicating component adaptations, how they work, and explain potential code adjustments. The tests directory contains all the prototypes in .json format and a Python script for their automated conversion. The Vue projects generated are available at [figma2vuejs-projects](#).

For a more nuanced comparison of the quality of code generated with Anima and FIGMA2VUEJS, we will focus on a prototype based on a movies streaming application (Figure 6).

TABLE III
CODE SIZE, COMPLEXITY AND DUPLICATION

Metric	Anima	FIGMA2VUEJS
Lines of Code (LOC)	3,924	10,081
LOC (no duplication)	3,924	9,772
Functions	1	132
Cyclomatic Complexity	2	143
Duplication Density	0.0%	2.2%
Duplicated Lines	0	309
Duplicated Blocks	0	28
Comment Lines	1	6

As illustrated in Table III, the Anima approach generates cleaner code with no duplicated lines or blocks of code, resulting in 0% duplication density, compared to the 2% in our tool. This indicates a need for further refinement in order to reduce code redundancy and produce higher maintainability. Even though the Vue project generated with FIGMA2VUEJS has more duplications, it presents a higher number of functions

TABLE IV
CODE FEATURE COMPARISON - ANIMA VS FIGMA2VUEJS

Code Feature	FIGMA2VUEJS	Anima
Page navigation	✓ (scripts)	✓ (templates)
Props	✓ (only #data components)	✓
Signal emissions	✓ (overlay open & close)	×
Conditional rendering	✓	×
Vue hooks	✓	×
Event handlers	✓	×
Data binding	✓ (one & two-way)	✓ (one-way)
Loop rendering (v-for)	✓	×
Dynamic components	✓ (variant components)	×
Validation Methods	✓ (form component)	×
Toast Service	✓ (using Pinia Store)	×

TABLE V
SECURITY, RELIABILITY, AND MAINTAINABILITY

Metric	Anima	FIGMA2VUEJS
Security Rating	A	A
Bugs (Reliability)	1	0
Reliability Rating	C	A
Code Smells	3	0
Technical Debt	15 min	0 min
Maintainability Rating	A	A

(132 vs 1 function) and a higher number of lines without duplication (9,772 vs 3,924 lines). Furthermore, the code produced by our tool achieved a higher cyclomatic complexity score (143) compared to the Anima code. This metric quantifies the number of linearly independent paths through a program's source code. However, that value is the result of functionally richer and scalable code, as demonstrated in Table IV.

With regard to the reliability metric, the Anima code shows one minor bug related to the lack of a *lang* attribute. This is a medium-category issue that requires a short time of 2 minutes. Another issue with the Anima code was related to the redundant *alt* attribute values in *img* elements, which represents a technical debt of 15 minutes. Meanwhile, in our approach, no code smells were detected, and there was no technical debt, which results in good codebase maintainability (Table V).

VI. CONCLUSION

Despite Anima generating a cleaner and simpler codebase, it lacks functional behavior. The Vue2 solution of Anima is best suited for small-scale projects with little associated logic and complexity. However, both the Anima plugin and FIGMA2VUEJS proved to be good solutions for achieving overall visual fidelity, as evidenced by the metrics results.

Our approach, which applies current Vue3 standards and architecture with the Options + Composition API, is more favourable for large-scale projects and professional maintenance. It also supports modular reactivity, state management with Pinia, and a more advanced logical composition. Additionally, our tool ensures a more precise separation between components and views by generating component files only for nodes of type "component" on the *artboard*, unlike Anima, which places all element nodes in the components folder, reducing developer control. Furthermore, Anima's strategy,

based on a centralised JSON data file passed via props, may ease API integration but could affect performance when components primarily handle static data.

To improve our tool, further work could focus on reducing duplicated code and improving visual accuracy between the prototypes and the generated web UIs. Reducing visual discrepancies will require mitigating the approximation errors in the grid system allocation. Although the overlapping of shapes of addressed, some visual accuracy was affected by the nature of the *clip-path* property. This issue could be mitigated by integrating components corresponding to specific overlaps of shapes into our design system, following a component-based prototyping approach. However, as we aimed to provide design flexibility, we opted for a generic solution capable of handling both shape nodes used in isolation or visually combined with other elements. Nevertheless, there is an opportunity to extend the tool by integrating more components into the supported design system.

The tool could be adapted to generate code using only the Composition or Options API, providing developers greater flexibility to choose their preferred standard. Future work also requires empirical evaluation with developers to quantify development time gains. This evaluation should measure the time needed to implement/modify visual or logic tasks using three approaches: hand-code only vs Anima vs FIGMA2VUEJS.

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